EXPERIMENT-4

AIM: To determine total hardness of water using EDTA solution.

Apparatus required: Burette, pipette, conical flask, beaker, measuring flask and droppers.

Reagent required:

1. Standard hard water (N/50, 1 mg $CaCO_3$/ 1ml water i.e. 1000 ppm.)
2. EDTA solution (N/50)
3. Ammonium chloride, ammonium hydroxide buffer (pH – 10)
4. Eriochrome Black – T indicator solution.

Total hardness:

Theory:

Hardness in water is that characteristic, which "prevents the lathering of soap". This is due to presence in water of certain salts of calcium, magnesium and other heavy metals dissolved in it. A sample of hard water, when treated with soap does not produce lather, but on other hand forms a white scum or precipitate. This precipitate is formed, due to the formation of insoluble soaps of calcium and magnesium.

Thus, water which does not produce lather with soap solution readily, but forms a white curd, is called hard water. On the other hand, water which lathers easily on shaking with soap solution, is called soft water. Such water consequently does not contain dissolved calcium and magnesium salts in it.

Temporary or carbonate hardness: It is caused by the presence of dissolved bicarbonates of calcium, magnesium and other heavy metals and the carbonate of iron. Temporary hardness is mostly destroyed by mere boiling of water, when bicarbonates are decomposed, will produce insoluble carbonates or hydroxides, which are deposited as a crust at the bottom of vessel.

Permanent or non-carbonate hardness: It is due to the presence of chlorides and sulphates of calcium, magnesium, iron, and other heavy metals. Unlike temporary hardness, permanent hardness is not destroyed on boiling.

The degree of hardness of drinking water has been classified in terms of the equivalent $CaCO_3$ concentration as follows:

Soft	**0-60mg/L**
Medium	**60-120mg/L**
Hard	**120-180mg/L**
Very Hard	**>180mg/L**

In a hard water sample, the total hardness can be determined by titrating the Ca^{2+} and Mg^{2+} present in an aliquot of the sample with Na_2EDTA solution, using NH_4Cl-NH_4OH buffer solution of pH 10 and Eriochrome Black-T as the metal indicator.

Na_2H_2Y (Disodium EDTA solution) → $2Na^+ + H_2Y^-$

$Mg^{2+} + HD^{2-}$(blue) → MgD (wine red) + H^+

D (metal-indicator complex, wine red colour) + H_2Y^- →Y^-(metal EDTA complex colourless) + HD^-(blue colour) + H^+

Ethylenediamine tetra-acetic acid (EDTA) and its sodium salts form a chelated soluble complex when added to a solution of certain metal cations. If a small amount of a dye such as Eriochrome black T is added to an aqueous solution containing calcium and magnesium ions at a pH of 10 ± 0.1, the solution will become wine red. If EDTA is then added as a titrant, the calcium and magnesium will be complexed. After sufficient EDTA has been added to complex all the magnesium and calcium, the solution will turn from wine red to blue. This is the end point of the titration.

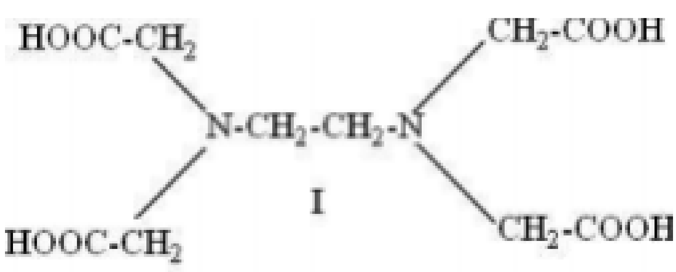
$$\text{Total hardness as } CaCO_3 \text{ (ppm)} = \frac{\text{Vol. of EDTA(mL)} \times 0.1 \times \text{molarity of EDTA} \times 10^6}{\text{Vol. of the sample (mL)}}$$

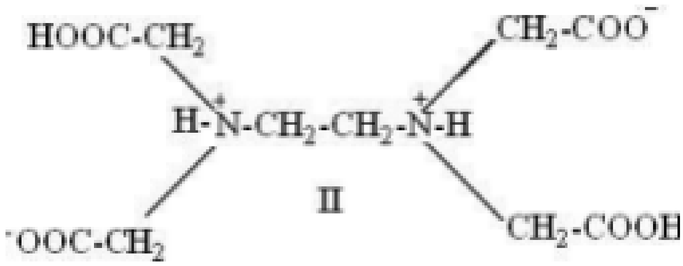
Principle of Complexometric Titration:

Hard water, which contains calcium and magnesium ions, forms a wine red coloured complex with the indicator, Erichrome Black-T, Ethylenediamine tetra acetic acid (EDTA) forms a colourless stable complex

with free metal ion like Ca, Mg. i.e., Metal +Indicator Metal indicator complex (wine red colour)

When EDTA is added from the burette, it extracts the metal ions from the metal ion indicator complex thereby releasing the free indicator. (The stability of metal ion- indicator complex is less than that of the metal ion- EDTA complex, and hence EDTA extracts metal ion form the ion indicator complex.)





Structure of EDTA

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The various EDTA species are abbreviated as H_4Y, H_3Y^-, H_2Y^{-2}, HY^{-3}, and Y^{-4}. The relative amounts of these species are a function of pH

At pH > 12

EDTA exist as Y^{-4}

pH > 8

EDTA exist as HY^{-3}

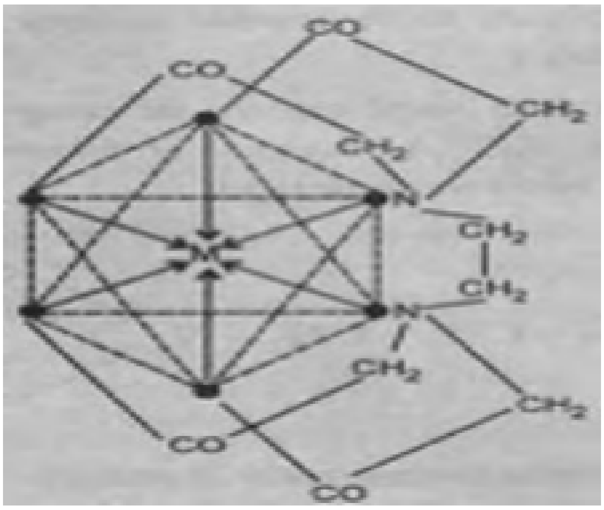
pH < 5 EDTA

exist as H_2Y^{-2}

The reactions take place at a pH = 1 and the buffer is made by ammonium chloride and ammonium solution. Complexometric titrations are particularly useful for determination of a mixture of different metal ions in solution. Ethylene diamine tetra acetic acid (EDTA), is a very important reagent for complex formation titrations. EDTA has been assigned the formula II in preference to I since it has been obtained from measurements of the dissociation constants that two hydrogen atoms are probably held in the form of zwitter ions. EDTA behaves as a dicarboxylic acid with two strongly acidic groups. For simplicity EDTA may be given the formula H_4Y, the disodium salt is therefore Na_2H_2Y and it has the complex forming ion H_2Y^{2-}in aqueous solution. The reactions with cations may be represented as;

$$M^{2+} + H_2Y_2^- \rightarrow MY^{2-} + 2H^+$$
$$M^{3+} + H_2Y_2^- \rightarrow MY^- + 2H^+$$
$$M^{4+} + H_2Y_2^- \rightarrow MY + 2H^+$$

One gram ion of the complex-forming ion H_2Y^{2-}reacts in all cases with one gram ion of the metal. EDTA forms complexes with metal ions in basic solutions. In acid-base titrations the end point is detected by a pH sensitive indicator. In the EDTA titration metal ion indicator is used to detect changes of pH. It is the negative logarithm of the free metal ion concentration, i.e., pH = - log $[M^{2+}]$. Metal ion complexes form complexes with specific metal ions. These differ in colour from the free indicator and a sudden colour change occurs at the end point.



Mechanism of action of indicator:

Eriochrome Black –T, a complex organic compound is added to hard water (pH=10 ± 0.1) it gives wine red color complex, with Ca^{2+} and Mg^{2+}ions of water sample. OH

OH

NaO_3S N=N O_2N

(1-(hydroxy 2-naphthhylazo) 6 nitro-2-napthol-4-sulphonate) **Erichrome Black T**

During an EDTA titration 2 complexes are formed:

i) M-EDTA complex and

ii) M-indicator complex. The metal-indicator complex must be less stable than the metal-indicator complex.

M-In + EDTA → M-EDTA + In

Erichrome black T is a metal ion indicator. In the pH range 7-11 the dye itself has a blue colour. In this pH range addition of metallic salts produces a brilliant change in colour from blue to red.

$$M^{2+} + HIn^{2-} \rightarrow MIn^{-} + H^{+}$$

(Blue) (Red)

This colour change can be obtained with the metal ions. As the EDTA solution is added, the concentration of the metal ion in the solution decreases due to the formation of metal-EDTA complex. At the end point no more free metal ions are present in the solution. At this stage, the free indicator is liberated and hence the colour changes from red to blue.

TABLE I.

S. No.	Volume of water sample taken(ml)	Burette reading		Volume of EDTA used(ml)
		Initial	Final	
1.	10	0.0	8.0	8.0
2.		8.0	16.0	8.0
3.		1.0	9.3	8.3
4.		9.3	17.4	8.1

Procedure

1. In a 250 ml conical flask pipette out 10 ml of prepared water sample.
2. Add 2ml of buffer solution and 1-2 drops of EBT
3. Add EDTA from the burette into the flask till the red colour of the solution changes to blue Colour.
4. Repeat the titration till to get three concordant readings

Calculation

Hard Water sample EDTA

N1V1 = N2V2

N1 = 0.02 X V2 / 10

N1 = ? .016

Strength of sample = N1 x 100 x 1000 ppm

Result:

Total hardness of the given water sample = ppm.